L01 — Pure/Mixed States and the Density Matrix

Lecture Metadata

Programme	Quantum Mechanics & Quantum Chemistry
Hierarchy	Series I > Module I.3 > Lecture L01
Series	Series I — Introductory Quantum Mechanics
Module	Module I.3 — Quantum Dynamics, Perturbation Theory & Physical Systems
Lecture	L01 — Pure/Mixed States and the Density Matrix
Duration	90 minutes
Active tracks	UG Grad Research
Active tiers	T1 HS T2 BegUG T3 AdvUG T4 MSc T5 PhD
Prerequisites	Module I.2 (L01–L12): bra-ket notation, operators, spectral decomposition, tensor products; basic linear algebra (eigenvalues, matrix exponential)
Preceding	I.2–L12: Tensor Products, Spin & Density Operators (introduction)
Following	L02: Bases — Discrete and Continuous (wavefunctions, Fourier transform)

Lecture summary. *This lecture introduces the density operator $\hat{\rho}$ as the fundamental descriptor of quantum states — pure, mixed, and reduced — replacing the single-ket picture that fails for open systems and statistical ensembles. Students learn to construct $\hat{\rho}$, verify its defining properties (Hermiticity, unit trace, positive semi-definiteness), compute expectation values via the trace, quantify mixedness via purity, and extract subsystem states via the partial trace. The density matrix unifies the entire Module I.3 treatment of dynamics and perturbation theory, and is the gateway to the open-system content of L10.*

Track routing.

- UG Tiers 1–3 receive: SCQU, CCQU, SPSU, CPSU, SPJU, CPJU
- Grad Tiers 4–5 receive: SCQG, CCQG, SPSG, CPSG, SPJG, CPJG
- Research Tiers 4–5 receive: WRQ×5, ORQ×5
- All artifact files are separate from this lecture body document.

1 Learning Objectives

Tier 1 (HS) — By the end of this lecture, you should be able to...

1. **Identify** why a single arrow (ket) is sometimes not enough to describe a quantum system, and give a physical example (e.g. a coin-flip mixture of spin-up and spin-down).
2. **Describe** the Bloch sphere picture: pure states sit on the surface ($|\mathbf{r}| = 1$) and mixed states live inside the sphere.
3. **Interpret** the diagonal entries of the density matrix as occupation probabilities, and the off-diagonal entries as coherences.
4. **Compute** (simple): given p_1, p_2 with $p_1 + p_2 = 1$, write down the density matrix for a classical mixture of two spin states.

Tier 2 (BegUG) — By the end of this lecture, you should be able to...

1. **Apply** the density-operator formula $\hat{\rho} = |\psi\rangle\langle\psi|$ for pure states and $\hat{\rho} = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ for mixtures to concrete 2-level examples.
2. **Calculate** expectation values via $\langle\hat{A}\rangle = \mathrm{Tr}(\hat{\rho}\hat{A})$ and verify agreement with the ket formula for pure states.
3. **Apply** the purity formula $\gamma = \mathrm{Tr}(\hat{\rho}^2)$ and classify a state as pure ($\gamma = 1$) or mixed ($\gamma < 1$).
4. **Translate** between the abstract ket description and its density-matrix representation in a chosen orthonormal basis.
5. **Calculate** the partial trace $\hat{\rho}_A = \mathrm{Tr}_B(\hat{\rho}_{AB})$ for simple $2 \otimes 2$ product states.

Tier 3 (AdvUG) — By the end of this lecture, you should be able to...

1. **Prove** that $\mathrm{Tr}(\hat{\rho}^2) \leq 1$ using the spectral decomposition of $\hat{\rho}$, with equality if and only if $\hat{\rho}$ is rank-1.
2. **Derive** the Bloch-vector parameterisation $\hat{\rho} = \frac{1}{2}(\hat{\mathbb{1}} + \mathbf{r} \cdot \hat{\boldsymbol{\sigma}})$ and verify all three defining properties.
3. **Construct** the partial trace from the tensor-product structure and verify it yields a valid density operator.
4. **Verify** that $\langle\hat{A}\rangle = \mathrm{Tr}(\hat{\rho}\hat{A})$ is basis-independent (trace cyclic property).
5. **Prove** global-phase invariance: $\hat{\rho}$ is unchanged by $|\psi\rangle \to e^{i\theta}|\psi\rangle$.

Tier 4 (MSc) — By the end of this lecture, you should be able to...

1. **Analyse** the purification of a mixed state: given $\hat{\rho}_A$, construct $|\Psi\rangle_{AB}$ such that $\mathrm{Tr}_B(|\Psi\rangle\langle\Psi|) = \hat{\rho}_A$.
2. **Connect** the Schmidt decomposition to the eigenstructure of reduced density matrices and to entanglement entropy.
3. **Generalise** the purity inequality to the sub-additivity of von Neumann entropy: $S(\hat{\rho}_{AB}) \leq S(\hat{\rho}_A) + S(\hat{\rho}_B)$.
4. **Reformulate** the partial trace in the Kraus-operator language and identify it as a special completely positive map.

Tier 5 (PhD) — By the end of this lecture, you should be able to...

1. **Formalise** density operators on infinite-dimensional Hilbert spaces as trace-class, positive, unit-trace operators in $\mathcal{T}_1(\mathcal{H})$.
2. **Classify** the convex set of density operators rigorously: extreme points are rank-1 projectors; the set is not a simplex for $d \geq 2$.
3. **Extend** the von Neumann entropy to the infinite-dimensional setting and state conditions for its finiteness.
4. **Critique** the Kraus representation theorem: state the GKSL theorem precisely and identify when a quantum channel is a partial trace.
5. **Classify** completely positive trace-preserving (CPTP) maps and their relationship to the Stinespring dilation theorem.

2 90-Minute Pacing Plan

Time	Topic / Activity	Tier depth note
0–15 min	**Motivation: why $\|\psi\rangle$ is not always enough.** Warm-up: Stern–Gerlach ensemble thought experiment. Distinguish (i) pure state, (ii) classical mixture, (iii) open-system subsystem. Introduce $\hat{\rho}$ as the unified descriptor.	T1: coin-flip analogy; T2: ket notation review; T3–T5: ray argument (global phase already absorbed)
15–40 min	**L01.3 Constructing $\hat{\rho}$.** Definitions: pure state $\hat{\rho} = \|\psi\rangle\langle\psi\|$; mixture $\hat{\rho} = \sum_i p_i \|\psi_i\rangle\langle\psi_i\|$. Three properties: (i) Hermitian, (ii) unit trace, (iii) positive semi-definite. Worked example: spin-$\frac{1}{2}$ mixture.	T2: matrix computation; T3: prove properties; T4: convexity of density operators; T5: trace-class operators
40–60 min	**L01.4 Expectation values and purity.** $\langle\hat{A}\rangle = \mathrm{Tr}(\hat{\rho}\hat{A})$; basis independence. Purity $\gamma = \mathrm{Tr}(\hat{\rho}^2) \in [1/d, 1]$; Bloch-sphere picture for $d = 2$. Worked example: compute $\langle\hat{\sigma}_z\rangle$ for a mixed qubit.	T1: Bloch sphere visual; T2–T3: purity calculation; T4: linear entropy $S_L = 1 - \gamma$; T5: von Neumann entropy preview
60–80 min	**L01.5 Partial trace and reduced density matrices.** Bipartite Hilbert space $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$. Definition and computation of $\hat{\rho}_A = \mathrm{Tr}_B(\hat{\rho}_{AB})$. Worked example: Bell state $\|\Phi^+\rangle$; partial trace is maximally mixed.	T2–T3: 2×2 computation; T4: Schmidt decomposition; T5: CPTP map interpretation
80–90 min	**Synthesis and connections.** Forward link to L02 (bases), L05 (time evolution preserves $\hat{\rho}$), L10 (Lindblad for open systems). Summary: the density operator is the object that time-evolves under the von Neumann equation $i\hbar\partial_t\hat{\rho} = [\hat{H}, \hat{\rho}]$. Q&A and tier-split problem preview.	All tiers: unifying picture; T4–T5: Liouville–von Neumann equation

3 Lecture Body

3.1 L01.3 — Motivation: When Is a Ket Not Enough?

Main result. A single ket $|\psi\rangle \in \mathcal{H}$ describes only *pure* quantum states. Three physically distinct situations require a more general description: (i) statistical ensembles of preparation procedures, (ii) quantum subsystems of entangled composite systems, and (iii) open systems coupled to unobserved environments. The density operator $\hat{\rho}$ handles all three cases uniformly.

Three Motivating Scenarios

Scenario 1: Statistical ensemble. A preparation machine selects $|\uparrow\rangle$ with probability p and $|\downarrow\rangle$ with probability $1-p$. No single ket encodes this classical uncertainty, yet all predictions follow from $\hat{\rho} = p|\uparrow\rangle\langle\uparrow| + (1-p)|\downarrow\rangle\langle\downarrow|$.

Scenario 2: Subsystem of entangled pair. Consider two qubits in the Bell state $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Qubit A alone has no valid ket description: its predictions are captured only by $\hat{\rho}_A = \mathrm{Tr}_B(|\Phi^+\rangle\langle\Phi^+|) = \frac{1}{2}\hat{\mathbb{1}}$.

Scenario 3: Open system. An atom interacts with a photon field. After tracing over field degrees of freedom, the atom's state is mixed, and its evolution is generated not by the Schrödinger equation but by a master equation for $\hat{\rho}$ (see L10).

Instructor note: *Draw the Bloch sphere on the board: T1 students focus on the sphere picture; T2–T3 on computing $\hat{\rho}$; T4–T5 on the convex geometry of the state space.*

3.2 L01.4 — Constructing the Density Operator

Main result. Any valid quantum state is described by a *density operator* $\hat{\rho}$ satisfying three conditions: (i) $\hat{\rho}^\dagger = \hat{\rho}$ (Hermitian), (ii) $\mathrm{Tr}(\hat{\rho}) = 1$ (unit trace), (iii) $\hat{\rho} \geq 0$ (positive semi-definite). Conversely, any operator satisfying these three conditions is a valid density operator.

Definitions

Pure state: $\hat{\rho} = |\psi\rangle\langle\psi|$ (rank-1 projector; $\hat{\rho}^2 = \hat{\rho}$)

Statistical mixture: $\hat{\rho} = \sum_i p_i\,|\psi_i\rangle\langle\psi_i|, \quad p_i \geq 0, \quad \sum_i p_i = 1$

Reduced density operator: $\hat{\rho}_A = \mathrm{Tr}_B(\hat{\rho}_{AB}) = \sum_j \big(\hat{\mathbb{1}}_A \otimes \langle j|_B\big)\hat{\rho}_{AB}\big(\hat{\mathbb{1}}_A \otimes |j\rangle_B\big)$

Expectation value: $\langle\hat{A}\rangle = \mathrm{Tr}(\hat{\rho}\hat{A})$ (basis-independent by cyclicity of trace)

Core Derivation: Proof of the Three Properties

Tier 3 **Tier 4** **Tier 5**

For the mixture $\hat{\rho} = \sum_i p_i|\psi_i\rangle\langle\psi_i|$ with $p_i \geq 0$, $\sum_i p_i = 1$:

1. **Hermitian:** $\hat{\rho}^\dagger = \sum_i p_i(|\psi_i\rangle\langle\psi_i|)^\dagger = \sum_i p_i|\psi_i\rangle\langle\psi_i| = \hat{\rho}$. ✓
2. **Unit trace:** $\mathrm{Tr}(\hat{\rho}) = \sum_i p_i\ \mathrm{Tr}(|\psi_i\rangle\langle\psi_i|) = \sum_i p_i\langle\psi_i|\psi_i\rangle = \sum_i p_i = 1$. ✓
3. **Positive semi-definite:** For any $|\phi\rangle$, $\langle\phi|\hat{\rho}|\phi\rangle = \sum_i p_i|\langle\phi|\psi_i\rangle|^2 \geq 0$, since each $p_i \geq 0$ and $|\langle\phi|\psi_i\rangle|^2 \geq 0$. ✓

Worked Example 1: Spin-$\frac{1}{2}$ mixture.
A source produces $|\uparrow\rangle$ with probability $\frac{3}{4}$ and $|\downarrow\rangle$ with probability $\frac{1}{4}$.

In the $\{|\uparrow\rangle, |\downarrow\rangle\}$ basis:

$$\hat{\rho} = \frac{3}{4}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{4}\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3/4 & 0 \\ 0 & 1/4 \end{pmatrix}. \tag{1}$$

Check: $\mathrm{Tr}(\hat{\rho}) = 1\checkmark$; $\hat{\rho} \geq 0$ (both eigenvalues positive)$\checkmark$; $\hat{\rho}^\dagger = \hat{\rho}\checkmark$.

3.3 L01.5 — Purity and the Bloch Sphere

Main result. The *purity* $\gamma = \mathrm{Tr}(\hat{\rho}^2)$ lies in $[1/d, 1]$ for a d-dimensional Hilbert space. $\gamma = 1$ if and only if $\hat{\rho}$ is a pure state; $\gamma = 1/d$ for the maximally mixed state $\hat{\rho} = \hat{\mathbb{1}}/d$.

Proof: $\mathrm{Tr}(\hat{\rho}^2) \leq 1$

Tier 3

By spectral decomposition, $\hat{\rho} = \sum_k \lambda_k |k\rangle\langle k|$ with $\lambda_k \geq 0$, $\sum_k \lambda_k = 1$. Then $\mathrm{Tr}(\hat{\rho}^2) = \sum_k \lambda_k^2$. By the Cauchy–Schwarz inequality (or by convexity of $t \mapsto t^2$):

$$\sum_k \lambda_k^2 \leq \left(\sum_k \lambda_k\right)^2 = 1. \tag{2}$$

Equality holds iff all probability is concentrated on one eigenvalue, i.e. $\hat{\rho} = |k_0\rangle\langle k_0|$ is rank-1 (pure state).

Bloch Sphere (qubit, $d = 2$)

Tier 2 **Tier 3**

Any qubit density operator can be written

$$\hat{\rho} = \tfrac{1}{2}(\hat{\mathbb{1}} + \mathbf{r} \cdot \hat{\boldsymbol{\sigma}}), \quad \mathbf{r} = (r_x, r_y, r_z) \in \mathbb{R}^3, \tag{3}$$

where $\hat{\boldsymbol{\sigma}} = (\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z)$ are the Pauli matrices. The conditions $\hat{\rho} \geq 0$ and $\mathrm{Tr}(\hat{\rho}) = 1$ are equivalent to $|\mathbf{r}| \leq 1$. Purity: $\gamma = \frac{1}{2}(1 + |\mathbf{r}|^2)$. Pure states: $|\mathbf{r}| = 1$ (Bloch sphere surface). Maximally mixed state: $\mathbf{r} = 0$, $\gamma = \frac{1}{2}$.

Worked Example 2: Expectation value via trace.
For the state in Example 1 and $\hat{A} = \hat{\sigma}_z = \mathrm{diag}(1, -1)$:

$$\begin{aligned} \langle\hat{\sigma}_z\rangle = \mathrm{Tr}(\hat{\rho}\hat{\sigma}_z) &= \mathrm{Tr}\begin{pmatrix} 3/4 & 0 \\ 0 & 1/4 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \mathrm{Tr}\begin{pmatrix} 3/4 & 0 \\ 0 & -1/4 \end{pmatrix} \\ &= \tfrac{3}{4} - \tfrac{1}{4} = \tfrac{1}{2}. \end{aligned} \tag{4}$$

Purity: $\gamma = \mathrm{Tr}(\hat{\rho}^2) = (3/4)^2 + (1/4)^2 = 9/16 + 1/16 = 10/16 = 5/8 < 1$ (mixed).

3.4 L01.6 — Partial Trace and Reduced Density Operators

Main result. For a composite system AB with state $\hat{\rho}_{AB} \in \mathcal{B}(\mathcal{H}_A \otimes \mathcal{H}_B)$, the *reduced density operator* of subsystem A is $\hat{\rho}_A = \mathrm{Tr}_B(\hat{\rho}_{AB})$. This is the unique operator such that $\mathrm{Tr}(\hat{\rho}_A \hat{A}) = \mathrm{Tr}(\hat{\rho}_{AB}(\hat{A} \otimes \hat{\mathbb{1}}_B))$ for all $\hat{A}$ on $\mathcal{H}_A$.

Computation

Choose an orthonormal basis $\{|j\rangle_B\}$ for $\mathcal{H}_B$. Then:

$$\hat{\rho}_A = \mathrm{Tr}_B(\hat{\rho}_{AB}) = \sum_j \left(\hat{\mathbb{1}}_A \otimes {}_B\langle j|\right) \hat{\rho}_{AB} \left(\hat{\mathbb{1}}_A \otimes |j\rangle_B\right). \tag{5}$$

Worked Example 3: Bell state partial trace.
$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Full state: $\hat{\rho}_{AB} = |\Phi^+\rangle\langle\Phi^+|$.
Using basis $\{|0\rangle_B, |1\rangle_B\}$:

$$\begin{aligned} \hat{\rho}_A &= {}_B\langle 0|\hat{\rho}_{AB}|0\rangle_B + {}_B\langle 1|\hat{\rho}_{AB}|1\rangle_B \\ &= \tfrac{1}{2}|0\rangle_A\langle 0| + \tfrac{1}{2}|1\rangle_A\langle 1| = \tfrac{1}{2}\hat{\mathbb{1}}_A. \end{aligned} \tag{6}$$

Interpretation: Qubit A is maximally mixed ($\gamma_A = 1/2$) even though the global two-qubit state is pure — a signature of entanglement.

von Neumann Entropy and Entanglement

Tier 4 **Tier 5**

$$S(\hat{\rho}_A) = -\mathrm{Tr}(\hat{\rho}_A \ln \hat{\rho}_A) = -\sum_k \lambda_k \ln \lambda_k \geq 0. \tag{7}$$

For the Bell state: $S(\hat{\rho}_A) = \ln 2$ (maximum for a qubit). For a product state $\hat{\rho}_{AB} = \hat{\rho}_A \otimes \hat{\rho}_B$: $S(\hat{\rho}_A)$ is the entropy of $\hat{\rho}_A$ alone (no entanglement contribution).

Purification

Tier 4

Given any $\hat{\rho}_A$ on $\mathcal{H}_A$ with spectral decomposition $\hat{\rho}_A = \sum_k \lambda_k |k\rangle_A\langle k|$, define the purification

$$|\Psi\rangle_{AB} = \sum_k \sqrt{\lambda_k}|k\rangle_A|k\rangle_B \in \mathcal{H}_A \otimes \mathcal{H}_B. \tag{8}$$

Then $\mathrm{Tr}_B(|\Psi\rangle\langle\Psi|) = \hat{\rho}_A$ by construction. This is the Schmidt decomposition of $|\Psi\rangle$.

4 Connections

Backward connections.

- **I.2–L12 (Tensor Products & Density Operators intro):** The present lecture fully develops the density matrix formalism introduced in the final lecture of Module I.2.
- **I.2–L04 (Eigenkets & Spectral Decomposition):** The spectral decomposition $\hat{\rho} = \sum_k \lambda_k |k\rangle\langle k|$ requires fluency with eigenket expansions.
- **I.2–L01 (Why Dirac Notation):** The ray picture established there (states as equivalence classes $|\psi\rangle \sim e^{i\theta}|\psi\rangle$) motivates the density operator as the object that absorbs the global phase.

Forward connections within Module I.3.

- **L02 (Bases):** The partial trace and expectation values are computed in specific bases; continuous-basis analogues appear there.
- **L05 (Time Evolution I):** The Liouville–von Neumann equation $i\hbar\,\partial_t\hat{\rho} = [\hat{H}, \hat{\rho}]$ shows how $\hat{\rho}$ time-evolves under a unitary; purity is conserved.
- **L07 (Heisenberg Picture):** $\langle \hat{A} \rangle = \mathrm{Tr}(\hat{\rho}\hat{A}_H(t))$ works in either picture; the density operator remains the state descriptor.
- **L10 (Open Systems, Lindblad):** The Lindblad master equation governs $\hat{\rho}$ for open systems; purity decreases with time as a signature of decoherence.

Cross-module connections.

- **Series II (Many-Body):** The one-body reduced density matrix $\hat{\gamma}^{(1)}$ in Module II.4 is the many-body analogue; its eigenvalues are natural occupation numbers.
- **Series III (Quantum Chemistry):** The electron density $\rho(\mathbf{r}) = \mathrm{Tr}(\hat{\rho}\,\hat{n}(\mathbf{r}))$ underpins DFT (Module III.3); the density matrix is the quantum-chemical tool for open-shell and excited states.

5 Reading and Resources

UNDERGRADUATE TRACK — Reading (Tiers 1–3)

[1] T1 **Penrose, R.** *The Emperor's New Mind*, Ch. 5–6. Oxford University Press, 1989. ★
Philosophical and pictorial introduction to quantum state description. Good pre-lecture reading for T1 students; connects wave-function collapse intuition to the density-matrix picture before formalism.

[2] T2 **Griffiths, D. J. & Schroeter, D. F.** *Introduction to Quantum Mechanics*, 3rd ed., § 3.5–3.6. Cambridge University Press, 2018. ★★★
Section 3.5 introduces the density matrix in the wave-mechanics language; Section 3.6 covers measurement and mixed states. Directly supports SPSU Part A problems on purity and expectation values.

[3] T2 **Townsend, J. S.** *A Modern Approach to Quantum Mechanics*, 2nd ed., Ch. 7. University Science Books, 2012. ★★
Chapter 7 develops density matrices for spin systems using the bra-ket formalism from Chapter 1 — a natural continuation of the Module I.2 reading. Supports SPSU Part B interpretation problems.

[4] T3 **Sakurai, J. J. & Napolitano, J.** *Modern Quantum Mechanics*, 3rd ed., § 3.4–3.6. Cambridge University Press, 2020. ★★★
The primary T3 reference. Section 3.4 (density operator), 3.5 (ensembles), 3.6 (time evolution of $\hat{\rho}$). Read alongside the CPSU proof problems on purity and the Bloch-vector derivation.

[5] T3 **von Neumann, J.** *Mathematical Foundations of Quantum Mechanics*, Ch. IV. Princeton University Press, 1955. ★★ **[M]**
The original introduction of the density matrix ("statistical operator"). Chapter IV is accessible at T3; provides historical context for WRQ 2.

[6] T3 **Fano, U.** Description of states in quantum mechanics by density matrix and operator techniques. *Rev. Mod. Phys.* **29**, 74–93 (1957). ★★ **[P]**
Classic pedagogical review introducing the density matrix for atomic and optical systems. Directly relevant to CPSU Problem 4 (ensemble representation) and WRQ 1.

GRADUATE TRACK — Reading (Tiers 4–5)

[7] T4 **Weinberg, S.** *Lectures on Quantum Mechanics*, 2nd ed., Ch. 3. Cambridge University Press, 2015. ★★★
Chapter 3 provides the cleanest graduate treatment of the density operator including mixed states, measurement, and decoherence. Supports CPSG Problem 1 (trace-class conditions) and SPJG Project 2.

[8] T4 **Ballentine, L. E.** *Quantum Mechanics: A Modern Development*, 2nd ed., Ch. 2. World Scientific, 2014. ★★★
Statistical interpretation of the density matrix and ensemble theory; detailed treatment of the relationship between mixed states and measurement. Supports CCQG and WRQ 3–4.

[9] **T4** **Nielsen, M. A. & Chuang, I. L.** *Quantum Computation and Quantum Information*, Ch. 2. Cambridge University Press, 2010. ★★★
Sections 2.4 (density operators) and 2.5 (Schmidt decomposition and purification) are the clearest modern treatment at the T4 level. Directly supports CPSG 2–3, SPJG 2–3, and WRQ 5.

[10] **T5** **Reed, M. & Simon, B.** *Methods of Modern Mathematical Physics*, Vol. I, Ch. VI. Academic Press, 1972. ★★★ **[M]**
Trace-class operators ($\mathcal{T}_1(\mathcal{H})$), their norms, and density operators as positive trace-class operators with unit trace. Essential for CPSG Problem 5 and CPJG Projects 1, 4.

[11] **T5** **Kraus, K.** *States, Effects, and Operations*, Springer Lecture Notes 190, 1983. ★★ **[M]**
The Kraus representation theorem and CPTP maps; shows the partial trace as the prototypical quantum channel. Supports CPJG Projects 3, 5 and ORQ 3.

[12] **T5** **Lindblad, G.** On the generators of quantum dynamical semigroups. *Commun. Math. Phys.* **48**, 119–130 (1976). ★★★ **[P]**
The GKSL master equation derived from complete positivity. The T5 anchor for ORQ 4–5. Preview for L10.

[13] **T5** **Gorini, V., Kossakowski, A. & Sudarshan, E. C. G.** Completely positive dynamical semigroups of n-level systems. *J. Math. Phys.* **17**, 821–825 (1976). ★★★ **[P]**
Independent derivation of the Lindblad form (GKSL). Complements Lindblad (1976); both papers together are the definitive reference for CPJG Project 5 and ORQ 4.

RESEARCH TRACK — Reading (Tiers 4–5)

[14] **T5** **Holevo, A. S.** *Quantum Systems, Channels, Information.* De Gruyter, 2012. ★★ **[M]**
Information-theoretic approach to quantum channels and density operators. Background for ORQ 2–3 and CPJG Projects 3, 5.

[15] **T5** **Wilde, M. M.** *Quantum Information Theory*, 2nd ed., Ch. 3–5. Cambridge University Press, 2017. ★★
Modern treatment connecting density operators, quantum channels, and entropy; freely available on arXiv (1106.1445). Background for WRQ 4–5 and ORQ 2.